

# JAMES NESBIT

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## PROFESSIONAL EXPERIENCE

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### Attentive

New York, NY

*Staff Data Scientist*

*Jan 2026 – Present*

*Senior Data Scientist*

*Jun 2025 – Jan 2026*

AI-powered SMS & email marketing platform for e-commerce brands. Lead science across brand on-site, experimentation, and conversational AI pillars.

- Diagnosed a platform-wide sign-up popup serving gap for Android Facebook in-app-browser users and shipped the fix by enabling an existing feature flag — projected to add 3.1M SMS and 11.8M email subscribers and \$270M+ in attributable brand revenue in year one.
- Owned the science and full ML implementation for AI Grow, a contextual bandit for sign-up popup optimization; delivered ~13% SMS / 11% email subscriber lift and ~10% revenue lift over manual optimization across brands.
- Shipped the brand-facing experimentation platform as science lead, now live across 1,600+ brands and ~134K experiments per year plus company-wide holdouts; designed the analysis stack around variance reduction (CUPED) and always-valid inference to shorten experiments and reduce revenue lost on suboptimal variants.
- Launched Attentive's first customer-facing GenAI agent (Insights Agent) as tech lead: customers ask plain-English questions about their performance and get answers in seconds across revenue and conversions, campaigns, journeys, list growth, and billing; built on Snowflake Cortex Analyst.
- Defined sign-up popup health & performance metrics that re-centered 2026 pillar strategy and are now a P0 company priority; entitlement simulation quantified a \$125M+ annual revenue opportunity.

### Amazon

Seattle, WA

*Senior Applied Scientist* (from Economist I/II → Applied Scientist II)

*Mar 2021 – Jun 2025*

Teams: WW Returns & ReCommerce Planning; F3 Distribution Optimization & Grocery Innovation.

- Designed and shipped large-scale optimization for transportation / vehicle routing, replenishment, inventory, and store-layout planning across thousands of Amazon Fresh and Whole Foods locations — \$100M+ in cost savings.
- Architected a forecasting and supply-chain planning platform for reverse logistics, identifying a \$230M planning-accuracy entitlement; built interpretable, decomposable hierarchical time-series models for actuals-vs-plan diagnosis.
- Established a statistical forecast goal-planning framework for Finance stakeholders, combining baseline forecasts with simulated business initiatives to define goal trajectories and bridge gaps between actuals and plan.
- Built an enterprise MLOps / forecasting platform (automated training, dev→prod promotion, DAG-based hierarchical forecasting) and causal-inference tooling quantifying the downstream impact of returns-policy changes.
- Delivered production science across a Java / Python / SQL / React + AWS stack; led 40+ technical interviews for forecasting and science roles.

## TECHNICAL SKILLS

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**Programming:** Python, Java, R, C++, Rust, SQL

**Cloud & Tools:** AWS (CDK / Python CDK), Snowflake, dbt

**Expertise:** Causal Inference & Experimentation, Reinforcement Learning, Optimization & Operations Research, Time-Series Forecasting, ML Infrastructure / MLOps

## RESEARCH & EDUCATION

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**Ph.D. in Economics**, New York University

2015 – 2021

Advisors: Tim Christensen, José Luis Montiel Olea, Alfred Galichon, Quang Vuong.

### Publications

- “(Machine) Learning Parameter Regions” (with J. L. Montiel Olea). *Journal of Econometrics* 222(1), 2021, 716–744.
- “A Robust Machine Learning Algorithm for Text Analysis” (with S. Ke and J. L. Montiel Olea). *Quantitative Economics* 15(4), 2024, 939–970.
- “Short-Term Fluctuations in Incidental Happiness and Economic Decision-Making: Experimental Evidence from a Sports Bar” (with J. Kessler, A. McClellan, A. Schotter). *Experimental Economics* 25(1), 2022, 141–169.

**Working Papers:** “Text as Instruments.”